Q.1 {\text{In the integration of polar coordinates}} dxdy {\text{is replaced by}}

A. d

B. rdrd

C. drd

D. rdr

Q.2 {{\text{If a point }}p(r,\theta) {\text{in polar coordinate system, then }}r{\text{ is the distance of }}p{\text{ from the }}}

A. {{{Pole}}}

B. {{{Polar axis}}}

C. {{{None of these}}}

D. {{{Imaginary axis}}}

Q.3 {\rm{The}}\,{\rm{value}}\,{\rm{of}}\,{\rm{integral }}\int\limits_0^3 {\int\limits_0^2 {x^2 dxdy} } \,\,{\rm{is}}

A. 16

B. 4

C. 8

D. 12

Q.4 {\text{The}} \int\limits_0^1 {{e^{{y^2}}}dx} =

A. {e^{{y^2}}}

B. 0

C. e + 1

D. e - 1

Q.5 \begin{gathered} {\text{The }}\,\,{\text{position }}\,\,{\text{of }}\,\,{\text{the}}\,\,{\text{ limacon }}\,\,{\text{relative }}\,\,{\text{to}}\,\,{\text{ the}}\,\,{\text{ polar}}\,\,{\text{ axis}}\,\,{\text{ depends}}\,\,{\text{ on }}\,\,{\text{whether}}\,\,{\text{ _____ }}\,\, \hfill \\ {\text{appears}}\,\,{\text{ in}}\,\,{\text{ the}}\,\,{\text{ equation }}\,\,{\text{and}}\,\,{\text{ whether}}\,\,\, + {\text{ }}\,\,{\text{or}}\,\,{\text{ }} - \,\,{\text{occurs}}{\text{.}} \hfill \\ \end{gathered}

A. (b)\,\,\,

B. (a)\,\,\,

C. {{(d)}}\,\,\,{{Both}}\,\,{{(a)}}\,\,{{or}}\,\,{{(b)}}{.}

D. (c)\,\,\,

Q.6 The rose curve has 2n-equally spaced petals of loops if n is _____.

A. odd

B. even

Q.7 {\text{The graph of the equation }}r\, = \,a(1 - \cos \,\theta){\text{ is symmetric about - - - - - - - - - }}{\text{.}}

A. {{y - axis}}

B. {{initial line}}

C. {{pole}}

D. {{None of these}}{{.}}

Q.8 Expression for the double integral for the volume of a solid bounded above by z=8-x-y and below by rectangle R=0 \leqslant x \leqslant 4,\,5 \leqslant y \leqslant 8 is.....

A. V = _5^8 {_0^4 {(8 - x - y)dxdy} }

B. V = _0^4 {_5^8 {(8 - x - y)dxdy} }

C. V = _5^8 {_0^4 {(8 - x - y)dydx} }

D. V = _5^8 {_0^4 {(8 - x - y)dx} }

Q.9 {\text{While changing Cartesian Integral into Polar integral, we change }}x{\text{ by - - - - - - - - - and }}y{\text{ by - - - - - - - - - - - - }}{\text{.}}

A. Sec\,\, ,\,\,\,\,\,\,\,Co \,\,

B. Cos\,\, ,\,\,\,\,\,\,\,Sin\,\,

C. Sin\,\, ,\,\,\,\,\,\,\,Cos\,\,

D. Co \,\, ,\,\,\,\,\,\,\,Sec\,\,

Q.10 Can we evaluate the following integral in given order of integration {\kern 1pt} \int\limits_0^{\frac{1}{2}} {\int\limits_{2x}^1 {{e^{{y^2}}}} } dydx

A. Yes

B. No
